

Part -1 Plan To Produce Production Control

Plan to Produce



Dynamics 365

PREPARED BY SHAIK ABDUL HAMEED

Summary of Contents:

•Plan to Produce Overview:

- It outlines the **end-to-end production process**, from planning production orders to executing and controlling them.
- This includes **demand forecasting, master planning, material requirements planning (MRP), production scheduling, execution, and reporting.**

•Key Modules Involved:

- The document discusses integration with modules such as:
 - **Master Planning**
 - **Production Control**
 - **Inventory Management**
 - **Warehouse Management**
 - **Product Information Management**

Production Types:

- Covers different types of production supported by D365, such as:
 - **Discrete manufacturing**
 - **Process manufacturing**
 - **Lean manufacturing**

Main Features in D365 for Plan to Produce:

- Bill of Materials (BOM)
- Routes and operations
- Resource and capacity planning
- Job scheduling and execution
- Costing and performance analysis

Process Flow:

- The flow generally includes:
 - Forecasting and demand planning
 - Master planning (MRP)
 - Creation and scheduling of production orders
 - Release and execution of orders
 - Reporting production progress and variances
 - Closing and costing

Three types of production

Discrete Manufacturing

◆ Definition:

Discrete manufacturing is the production of distinct items that can be counted, touched, and seen—such as cars, appliances, or electronics. It typically involves assembly lines and workstations where components are combined into finished products.

◆ Key Characteristics:

- Products are made in **separate, individual units**.
- Often involves **Bill of Materials (BOMs)** and **Routes**.
- Production orders can be **made-to-stock, made-to-order, or engineer-to-order**.
- The process is **flexible**—it can be easily modified for new or custom products.

◆ Example:

Manufacturing of laptops, vehicles, or furniture.

◆ In D365:

- Use **Production Orders** to track and manage manufacturing.
- Configure **BOMs** for components and **Routes** for operations.
- Monitor **costing, execution, scrap, and production status** in real-time.
- Supports tracking by **serial or batch numbers**.

Process Manufacturing

◆ Definition:

Process manufacturing is used for **continuous or batch production** of products that cannot be easily disassembled—typically for chemicals, beverages, food, or pharmaceuticals.

◆ Key Characteristics:

- Products are **formulated**, not assembled.
- Often involves recipes or formulas rather than BOMs.
- Ingredients are usually measured by **weight or volume**.
- Emphasizes **batch control**, **yield management**, and **compliance** with quality and safety regulations.

◆ Example:

Manufacturing of paint, soda, shampoo, or medicine.

◆ In D365:

- Uses **Formulas** (instead of traditional BOMs).
- Tracks **Co-products** and **By-products** (e.g., waste material or secondary items).
- Manages **potency**, **shelf-life**, **lot tracking**, and **quality specifications**.
- Handles **batch orders**, **scaling**, and **variable output**.

Lean Manufacturing

◆ Definition:

Lean manufacturing focuses on **eliminating waste**, improving efficiency, and producing just what is needed, when it's needed. It uses **Kanban**, **pull-based systems**, and **continuous flow** rather than traditional orders.

◆ Key Characteristics:

- Based on **demand-driven (pull)** systems.
- Uses **Kanban cards** and **boards** to control work.
- Emphasizes **visual controls**, **cellular manufacturing**, and **small batch sizes**.
- Suitable for **repetitive production environments** with high stability.

◆ Example:

Automotive components production with a high focus on minimizing inventory and setup time.

◆ In D365:

- Uses **Kanban boards**, **activities**, and **rules** to manage tasks.
- Supports **event-driven** and **scheduled** Kanbans.
- Enables **production flow models** with process steps defined by work cells.
- Reduces reliance on production orders; focus is on **flow efficiency** and **real-time execution**.

Summary Comparison Table:

Feature	Discrete Manufacturing	Process Manufacturing	Lean Manufacturing
Output	Countable units	Bulk or batch-based	Flow or pull-based units
Input Structure	BOM + Routes	Formulas + Routes	Kanban + Production Flows
Tracking	Serial/Batch	Batch, Lot, Potency	Kanban ID, Activities
Planning Method	MRP/Production Orders	Batch Orders	Pull systems
Common Industries	Electronics, Machinery	Chemicals, Food, Pharma	Automotive, Repetitive
Focus	Flexibility	Compliance, Yield	Waste reduction, Flow

How to Configure Bill of Materials (BOM) in Microsoft Dynamics 365 Finance and Operations

Overview:

A **Bill of Materials (BOM)** is a structured list of raw materials, components, and sub-assemblies required to manufacture a finished product. In **D365 F&O**, BOMs help streamline production planning, inventory control, and cost management.

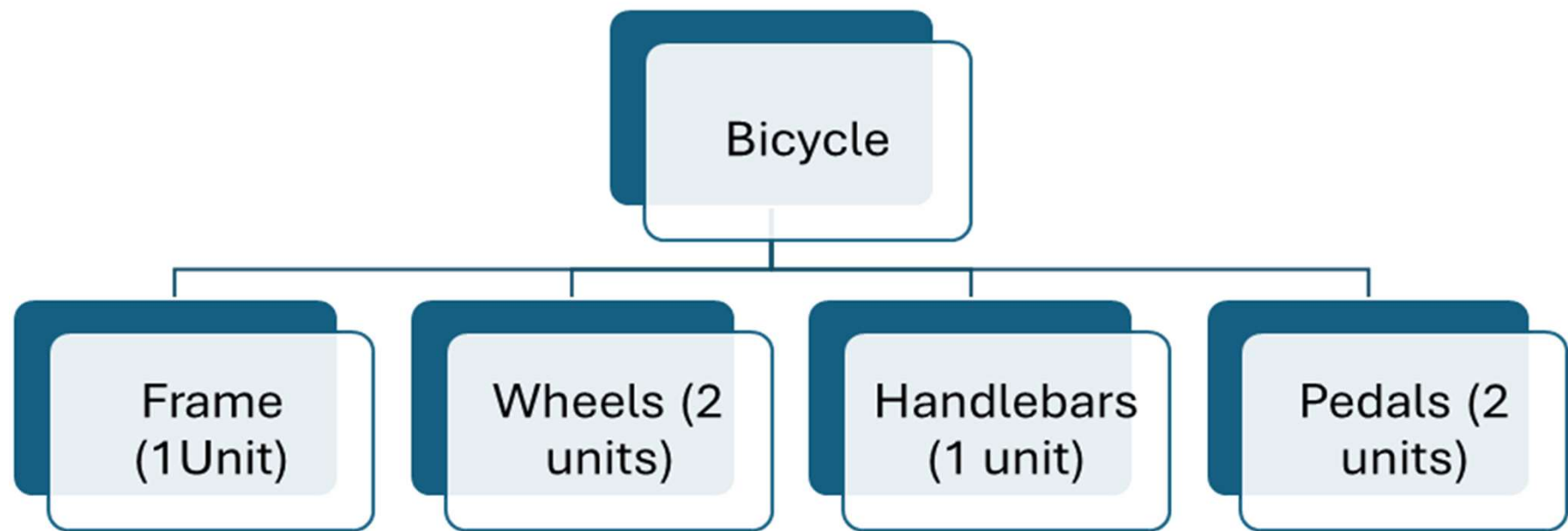
This guide will walk you through **creating a BOM** step by step.

Prerequisites:

Let's take an example of creating a **Bicycle**. By following the steps below, you can define a BOM for the bicycle:

- Finished Product:** Bicycle

- BOM Components:** Frame (1 unit) Wheels (2 units) Handlebars (1 unit) Pedals (2 units)



Step-by-Step Guide to Creating a BOM:

Step 1: Create a Finished Product (Bicycle)

Go to Product Information Management module > Products > Released Products > New > Define Basic Product
Details like (Product type, Product number, Product Name, Item model group, Item Group, Storage and tracking
dimension, UOM (Inventory, Purchase, Sales, BOM) etc. > Validate > Save

Finance and Operations

Product information management > Products > Released products

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Accounts receivable

Asset leasing

Asset management

Audit workbench

Benefits management

Budgeting

Cash and bank management

Common

Consolidations

Cost accounting

Cost management

Credit and collections

Engineering change management

Expense management

Fixed assets

General ledger

Global inventory accounting

Human resources

Inventory management

Landed cost

Leave and absence

Master planning

Released product details | Standard view

P-000020 : Bicycle

Summary by Copilot

General

IDENTIFICATION

Product type

Item

Product subtype

Product

Product service type

Not specified

Item number

P-000020

Product number

P-000020

Engineering product category de...

Product owner

FURTHER IDENTIFICATION

Product name

Bicycle

Search name

Bicycle

Description

ADMINISTRATION

Storage dimension group

Site&WH

Tracking dimension group

None

Item model group

FIFO

Reservation hierarchy

Counting reason code policy

Product lifecycle state

Advanced notes group

RELEASE

Release policy

READINESS

Product readiness policy

Purchase

Promote

Deliver

Sell

Foreign trade

Manage inventory

Engineer

BOM unit

ea

Constant scrap

Variable scrap

Costing level

0

Planning level

0

Cost calculation level

0

Phantom

No

Auto-report as finished

No

MEASUREMENT

Height

Width

Depth

Density

CALCULATION

Calculation group

STD

PRODUCTION

Production pool

Production group

Property

Arrival

No

Flushing principle

Start

Release to warehouse

Allow partial reservation

FORMULA PLANNING

Production type

BOM

Planning formula

Set the Production type to "BOM" for the Finished Product to activate the BOM section and enable BOM creation

Step 2: Create Sub-Assembly Products (Frame, Wheels, Handlebars, Pedals)

Go to Product Information Management module > Products > Released Products > New > Define Basic Product Details like (Product type, Product number, Product Name, Item model group, Item Group, Storage and tracking dimension, UOM (Inventory, Purchase, Sales, BOM) etc. > Validate > Save

Summary by Copilot ▼

P-000021

IDENTIFICATION	Product service type	Engineering product category de...	FURTHER IDENTIFICATION	ADMINISTRATION	Item model group	Product lifecycle state	RELEASE
Product type	Not specified		Product name	Storage dimension group	FIFO		Release policy
Item	Item number	Product owner	Frame	Site&WH	Reservation hierarchy	Advanced notes group	
Product subtype	P-000021		Search name	Tracking dimension group			READINESS
Product	Product number		Frame	None	Counting reason code policy		Product readiness policy
	P-000021		Description				

5.00

0.00

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

None

BOM unit ea	Costing level 0	Phantom ☐ No	Width 	CALCULATION	Production group 	Flushing principle Start	FORMULA PLANNING
Constant scrap 	Planning level 0	Auto-report as finished ☐ No	Depth 	Calculation group STD	Property 	Release to warehouse 	Production type None
Variable scrap 	Cost calculation level 0	MEASUREMENT	Density 	PRODUCTION	Arrival ☒ No	Allow partial reservation 	Planning formula
	Height 			Production pool 			

Set Production type to "None" for sub-assembly products

Finance and Operations

Product information management > Products > Released products

USMF

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Released product details | Standard view

P-000022 : Wheels

Summary by Copilot

General

IDENTIFICATION

Product type

Item

Product subtype

Product

Product service type

Not specified

Item number

P-000022

Product number

P-000022

Engineering product category de...

Product owner

FURTHER IDENTIFICATION

Product name

Wheels

Search name

Wheels

Description

ADMINISTRATION

Storage dimension group

Site&WH

Tracking dimension group

None

Item model group

FIFO

Reservation hierarchy

Counting reason code policy

Product lifecycle state

Advanced notes group

RELEASE

Release policy

READINESS

Product readiness policy

Purchase

Promote

Deliver

Sell

Foreign trade

Manage inventory

Engineer

BOM unit

ea

Constant scrap

Variable scrap

Costing level

0

Planning level

0

Cost calculation level

0

Phantom

No

Auto-report as finished

No

MEASUREMENT

Height

Width

Depth

Density

CALCULATION

Calculation group

STD

PRODUCTION

Production pool

Production group

Property

Arrival

No

Flushing principle

Start

Release to warehouse

Allow partial reservation

FORMULA PLANNING

Production type

None

Planning formula

Set Production type to "None" for sub-assembly products

Summary by Copilot ▼

Purchase	€3	--	4.00	▼
Promote				▼
Deliver				▼
Sell	€3	--	0.00	▼
Foreign trade		--	--	▼
Manage inventory	€3			▼

Engineer ea None ^

BOM unit ea	Costing level 0	Phantom ☐ No	Width 	CALCULATION Calculation group STD	Production group 	Flushing principle Start	FORMULA PLANNING Production type None
Constant scrap 	Planning level 0	Auto-report as finished ☐ No	Depth 	PRODUCTION Production pool 	Property 	Release to warehouse 	Planning formula
Variable scrap 	Cost calculation level 0	MEASUREMENT Height 	Density 		Arrival ☒ No	Allow partial reservation 	

 Set Production type to "None" for sub-assembly products

- ⇒ Set Production type to "None" for sub-assembly products

Finance and Operations

Product information management > Products > Released products

USMF

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Released product details | Standard view

P-000024 : Pedals

Summary by Copilot

General

P-000024

IDENTIFICATION	Product service type Not specified	Engineering product category de...	FURTHER IDENTIFICATION	ADMINISTRATION	Item model group FIFO	Product lifecycle state	RELEASE
Product type			Product name Pedals	Storage dimension group Site&WH	Reservation hierarchy	Advanced notes group	Release policy
Item	Item number P-000024	Product owner	Search name Pedals	Tracking dimension group None	Counting reason code policy		READINESS
Product subtype	Product number P-000024		Description				Product readiness policy
Product							

Purchase

ea | -- | 3.00

Promote

Deliver

Sell

ea | -- | 0.00

Foreign trade

-- | --

Manage inventory

ea

Engineer

ea | None

BOM unit ea	Costing level 0	Phantom ☒ No	Width	CALCULATION	Production group	Flushing principle Start	FORMULA PLANNING
Constant scrap	Planning level 0	Auto-report as finished ☒ No	Depth	Calculation group STD	Property	Release to warehouse	Production type None
Variable scrap	Cost calculation level 0	MEASUREMENT	Density	PRODUCTION	Arrival ☒ No	Allow partial reservation	Planning formula
		Height		Production pool			

Set Production type to "None" for sub-assembly products

Step 3: Navigate to the BOM Module

Go to Modules > Product information management > Select the Finished good product > Select the Engineer tab (Action Pane) > BOM > BOM Versions > New > BOM and BOM Version > Create BOM > Name (Give BOM Name) > Site > Ok

- Maintain BOM
- Copy BOM
- Bills of materials
- Check
- Product change
- Create case
- Associate with case
- All cases
- Add to case log

Filter

We didn't find anything to show here.

BOM versions | P-000020 : Bicycle | Standard view

BOM version

IDENTIFICATION	INVENTORY DIMENSIONS	PRODUCT REFERENCE	VALIDITY	To date	STATUS	Approved
BOM	Site	Item number	From qty. 0.00		Active	No
Name		Product name	From date	Engineering controlled	No	Approved by

Bill of materials lines

+ New Delete Edit Where-used BOM versions Display dimensions Product change Engineering change

Product number Item number Configuration Size Color Style Warehouse Resource c... Quantity Per series Unit Plan group Priority Configurati

We didn't find anything to show here.											
---------------------------------------	--	--	--	--	--	--	--	--	--	--	--

Create BOM

Parameters

Create a new BOM and BOM version.

Name
Bicycle BOM

Copy
No

Site
1

Select the site for which you are creating this BOM

OK Cancel

Step 4: Add BOM Components

In the **BOM lines** section, click **New** to add components > Select the **Item number** (sub-assemblies' products) > Define the **Quantity** required > Define Per series Qty. > **Unit of measure** (e.g., ea) > Repeat for all materials and sub-components

Finance and Operations

Product information management > Products > Released products

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Bill of materials

001772 : Bicycle BOM

Standard view

001772 : Bicycle BOM

Lines

Header

Bill of materials header

BOM

001772

Name

Bicycle BOM

Site

1

Item group

Audio

Approved by

Approved

No

Use configuration nomenclature

No

Bill of materials lines

+ New

Delete

Sorting

Where-used

BOM versions

Configuration

Display dimensions

Product change

Engineering change

	Engineerin...	Item number	Configuration	Size	Color	Style	Warehouse	Resource c...	Quantity	Per series	Unit	Plan group	Priority	Configuration group	It...	Product name
		P-000021					11	☐	1.0000	1	ea		0		🔗	Frame
		P-000022					11	☐	2.0000	1	ea		0		🔗	Wheels
	☒	P-000023	▼				11	☒	1.0000	1	ea	▼	0		▼	🔗 Handlebars
		P-000024					11	☐	2.0000	1	ea		0		🔗	Pedals

Select the quantity needed for one finished good

Per series" indicates the quantity of finished goods produced

Line details

General

Setup

Step consumption

Dimension

IDENTIFICATION

Configuration group

Item number

P-000023

Position

SUBCONTRACTOR

Vendor account

BOM

Line type

Item

Calculation

Yes

Sub-BOM or formula

Subroute

PROJECT

Set subproduction to Consumed

No

VALID

From

To

OPERATION

Oper. No.

End

No

Step 5: Approve and Activate the BOM

Click Approve to approve the BOM > Click Activate to make the BOM available for production.

Approve BOM or formula

Parameters

Approved by
000863

Name
Arvind Bharti

Approve the
BOM

OK Cancel

Maintain

Designer Approval Check Copy

Configuration

Configuration rules

Product change

Create case All cases Associate with case Add to case log

Bills of materials | 001772 : Bicycle BOM | Standard view

001772 : Bicycle BOM

Lines Header

Bill of materials header

BOM

001772

Name

Bicycle BOM

Site

1

Item group

Audio

Approved by

000863

Approved

Yes

Use configuration nomenclature

No

Bill of materials lines

+ New Delete Sorting Where-used BOM versions Configuration Display dimensions Product change Engineering change

	Engineerin...	Item number	Configuration	Size	Color	Style	Warehouse	Resource c...	Quantity	Per series	Unit	Plan group	Priority	Configuration group	It...	Product name
		P-000021					11	☐	1.0000	1	ea		0			Frame
		P-000022					11	☐	2.0000	1	ea		0			Wheels
☒	☐	P-000023					11	☐	1.0000	1	ea		0			Handlebars
		P-000024					11	☐	2.0000	1	ea		0			Pedals

Line details

General Setup Step consumption Dimension

IDENTIFICATION

Configuration group

Item number

P-000023

Position

SUBCONTRACTOR

Vendor account

BOM

Line type

Item

Calculation

Yes

Sub-BOM or formula

Subroute

PROJECT

Set subproduction to Consumed

No

VALID

From

To

OPERATION

Oper. No.

End

No

Finance and Operations

Operation completed

Save Delete + New BOM version Bill of materials Options

Maintain BOM

Copy BOM

Bills of materials

Check

Product change

Create case

Associate with case

All cases

Add to case log

Filter

P-000020

Bicycle

P-000020

BOM versions | P-000020 : Bicycle | Standard view

P-000020 : Bicycle

BOM version

IDENTIFICATION	PRODUCT REFERENCE	From date
BOM	Item number	
P-000020	P-000020	
Name	Product name	To date
Bicycle	Bicycle	
INVENTORY DIMENSIONS	VALIDITY	STATUS
Site	From qty.	Active
Site -1	1.00	No

Bill of materials lines

+ New Delete Edit Where-used BOM versions Display dimensions Product change

Item number	Configuration	Size	Color	Style
P-000021				
P-000022				

Check BOM or formula

Parameters

Check circularity



Yes

Circularity check strategy

Optimize for high complexity

Display principle

Active

BOM number

P-000020

INTERVAL

Search interval only



No

Search interval with lines only



No

From date

To date

Run in the background

Recurrence Alerts

Batch processing



No

Task description

Check BOM or formula

Batch group

Private



No

Critical Job



No

Monitoring category

OK

Cancel

▼

BOM versions -- Finance and

Plan to Produce Overview

(7) How to Configure Bill of

(8) Part-1: What is "Plan to P

(8) How to Configure Bill of

+

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📄

✕

← → ↻ 🔍 usnconeboxax1aos.cloud.onebox.dynamics.com/?cmp=m1&mi=EcoResProductDetailsExtendedGrid

★ ⬇️ Ⓢ ⋮

Finance and Operations

Search for a page

Operation completed

← ⋮ Save Delete + New ▼

BOM version

Bill of materials

Options 🔍

Maintain BOM

Copy BOM

Bills of materials

Check

Product change

Create case

Associate with case

All cases

Add to case log

Filter

P-000020

Bicycle

P-000020

BOM versions | P-000020 : Bicycle | Standard view ▼

P-000020 : Bicycle

BOM version

IDENTIFICATION

BOM

P-000020 ▼

Name

Bicycle

INVENTORY DIMENSIONS

Site

Site -1

PRODUCT REFERENCE

Item number

P-000020

Product name

Bicycle

VALIDITY

From qty.

1.00

From date

To date

STATUS

Active

No

Bill of materials lines

+ New ▼ Delete Edit Where-used BOM versions Display dimensions Product change ▼

Item number	Configuration	Size	Color	Style
P-000021				
P-000022				

Standard view ▼

Define recurrence

Start date

7/15/2025

Start time

09:30:27 PM

Time zone

(GMT+05:30) Chennai, Kolkata, Mumbai, Ne... ▼

☐ NO END DATE

☒ END AFTER:

☐ END BY:

1

7/15/2025

RECURRENCE PATTERN

☒ Minutes

☐ Hours

☐ Days

☐ Weeks

☐ Months

☐ Years

Repeat after specified number of minutes

Count

10

OK

Cancel

Windows Taskbar

9:31 PM 7/15/2025

Maintain BOM version

Approval

Deactivate

Calculation

 Filter

P-000020

Bicycle BOM

001772

BOM versions | P-000020 : Bicycle | Standard view 

P-000020 : Bicycle BOM

BOM version

001772 Bicycle BOM 

IDENTIFICATION	INVENTORY DIMENSIONS	PRODUCT REFERENCE	VALIDITY	To date	STATUS	Approved
BOM 001772	Site 1	Item number P-000020	From qty. 1.00		Active ☒ Yes	☒ Yes
Name Bicycle BOM		Product name Bicycle	From date 	Engineering controlled ☒ No		Approved by 000863

Bill of materials lines

   Where-used BOM versions Display dimensions Product change  Engineering change 

☐	Product number	Item number	Configuration	Size	Color	Style	Warehouse	Resource c...	Quantity	Per series	Unit	Plan group	Priority	Configuration group	It...	Product name
☒	P-000021	P-000021					11		1.0000	1	ea		0			Frame
☐	P-000022	P-000022					11		2.0000	1	ea		0			Wheels
☐	P-000023	P-000023					11		1.0000	1	ea		0			Handlebars
☐	P-000024	P-000024					11		2.0000	1	ea		0			Pedals

Conclusion:

BOM is created and activated, the product is ready for **production planning, inventory management, and cost tracking** in **Microsoft Dynamics 365 Finance and Operations**.

THANK YOU FOR THE SUPPORT GUYS